Greenlight Commute

Greenlight Commute: Executive Summary

The Problem

Universities are under rising pressure to measure and reduce their carbon footprints, yet most cannot say how much of that footprint comes from the way their own people get to campus. Commuting is typically the largest single component of a campus's Scope 3 emissions; in published campus inventories it commonly accounts for 20 to 40 percent of total emissions, and at commuter-heavy schools more. It is also the component hardest to measure.

The standard tool is an annual travel survey. Our interviews with eleven sustainability officers at institutions in our region confirmed what the literature suggests: response rates average 8 to 15 percent, answers describe a single typical day rather than actual behavior, and results arrive months late. One officer told us her office reports a commuting figure "we know is wrong but can't improve." Meanwhile, more than 1,100 US campuses have signed public climate commitments with reporting obligations, and the offices responsible are typically staffed by one to three people with no data engineers.

The Solution

Greenlight Commute is a mobile app paired with an advisory service.

For commuters, the app, used with opt-in consent, logs trips to and from campus, identifies the travel mode, and estimates the emissions of each trip. It then shows each user a personalized "switch plan": for example, replacing three of five weekly solo drives with a bus route that adds nine minutes to the trip and saves 11.4 kilograms of carbon dioxide per week. It matches carpool partners among enrolled users, surfaces transit and bike-share options, and gives modest rewards, such as campus-store credits funded by the university, for sustained mode shifts.

For sustainability offices, we provide an anonymized, aggregated dashboard showing emissions by mode, by origin zone and by population group, plus an annual advisory report formatted for climate-action reporting and a set of targeted intervention recommendations, such as where a new shuttle stop would remove the most solo-driving trips. Individual data is never shared with the institution. Location data is aggregated before leaving our servers, and users can delete their history at any time.

The distinctive element is that we deliver measurement and behavior change together. Survey vendors measure but do not change behavior; consumer carbon apps change behavior but give institutions no usable data. Our pilot shows the combination works.

Market

There are roughly 3,900 degree-granting institutions in the United States enrolling about 19 million students and employing several million faculty and staff. We focus first on the roughly 1,100 campuses with public climate commitments and a dedicated sustainability office. At our target price of $4.50 per enrolled commuter per year, and assuming 40 percent of enrolled populations commute, serving these campuses represents a serviceable market of about $38 million annually. Adding advisory work and extending to hospitals and large employers, which face similar reporting pressure, raises the opportunity above $200 million. Our initial beachhead is the 140 campuses within a day's travel of our university, where in-person advisory is cost-effective.

Business Model

Revenue comes from two streams. The first is an annual software license priced per enrolled commuter, ranging from $3.00 to $5.00 depending on volume, with a minimum annual fee of $6,000. The second is advisory services: the annual report and intervention analysis at $9,500 for a

standard package, and custom studies for transit-planning projects at $15,000 to $30,000. Universities can fund these from sustainability budgets, parking and transportation revenue, or green-fee funds, which several campuses already administer for exactly this purpose. Our commuter-facing app is free, so adoption is not constrained by individual willingness to pay. We estimate gross margin at 78 percent for licenses and 55 percent for advisory work, and a customer acquisition cost of about $4,000 per campus, recovered within the first contract year.

Competition

Campus transportation surveys offered by consulting firms are slow, costly at $15,000 to $40,000 per cycle, and low in response. Corporate commute-management platforms focus on parking and employee benefits rather than emissions reporting, and do not serve student populations. Consumer carbon-footprint apps lack institutional dashboards and rely on self-reported trips. Transit agencies collect ridership data but nothing on trips they do not carry. No competitor we found combines passive trip measurement, behavior-change recommendations and reporting designed for campus climate commitments. Our advantage is reinforced by our data: each enrolled campus improves our mode-detection accuracy and our estimates of the impact of each type of intervention.

Team

Priya Raghunathan, chief executive, is a senior in environmental economics who managed our university's student sustainability fund and led the commuter-survey redesign that raised campus response from 11 to 19 percent. Marcus Delacroix-Whitfield, chief technology officer, is a computer science senior who built the trip-detection engine and previously interned on a mobile data team, shipping a location-based feature used by 200,000 people. Tomasz Okonkwo-Nowak, head of partnerships, is a master's student in urban planning who spent two years working with the regional transit authority on service-gap analysis. Our faculty advisor in the sustainability institute has overseen three campus emissions inventories.

Milestones and Financials

We have formed Greenlight Commute LLC and defined the core product concept and data model. Our pilot on our own campus enrolled 1,840 commuters over one semester, 11 percent of the commuting population, and signed two paid agreements: a $6,000 annual license with our university's sustainability office and a $9,500 advisory engagement with a neighboring college. Among pilot participants who received switch plans, solo-driving trips fell 14 percent relative to a matched group over twelve weeks, avoiding an estimated 21 metric tons of carbon dioxide. Early projections: $15,500 in booked revenue this year, rising to $190,000 in year two with eight campuses and $720,000 in year three with twenty-five. We expect to reach breakeven during year three, with costs driven mainly by two engineers, a part-time analyst and cloud infrastructure.

What We Ask For

We are seeking a $25,000 prize or seed grant to fund our next twelve months: roughly $12,000 to harden the app's trip detection and privacy architecture, $8,000 to convert six campuses now in conversation from pilot discussions to signed contracts, and $5,000 for legal review of data-protection agreements. We would also value introductions to campus sustainability networks and mentorship from judges experienced in selling to higher education. With this support, we will have eight paying campuses and a repeatable sales process by the end of next year.